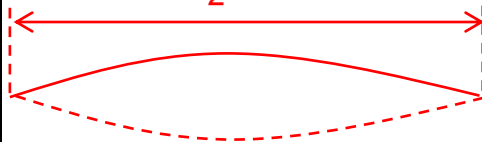


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D

$$\frac{\lambda}{2} = 1 \text{ m}$$

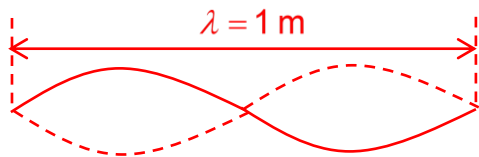


At the fundamental mode,

$$\frac{\lambda}{2} = 1 \text{ m}, \lambda = 2 \text{ m} \rightarrow \text{Option A is incorrect.}$$

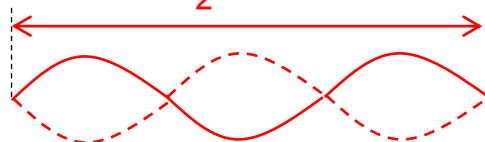
The mid-point of the rope is not stationary at fundamental mode. Option B is incorrect.

2nd Harmonic- $\lambda = 1 \text{ m}$

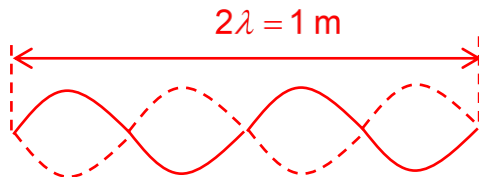


3rd Harmonic- $\lambda = 0.667 \text{ m}$

$$\frac{3\lambda}{2} = 1 \text{ m}$$



4th Harmonic- $\lambda = 0.5 \text{ m}$



5th Harmonic- $\lambda = 0.4 \text{ m}$

$$\frac{5\lambda}{2} = 1 \text{ m}$$



When the fundamental frequency is doubled, there are 3 nodes on the rope. Option C is incorrect.

At the 5th harmonic, the wavelength is 0.4 m. Option D is correct.

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B